C2 - A basic codec

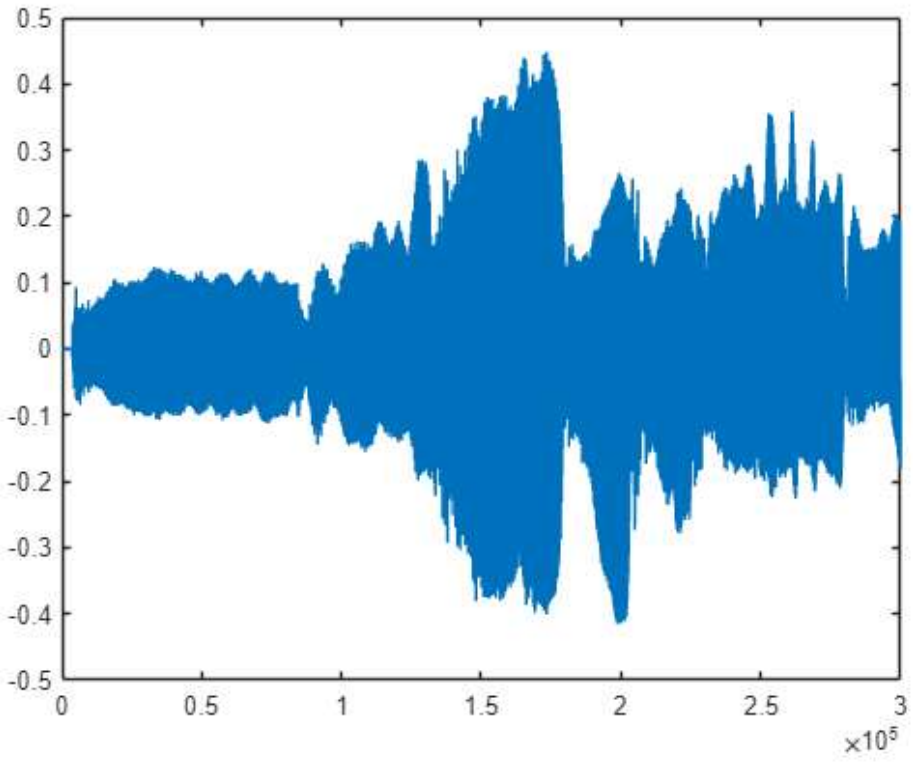
C2.1 - Preliminary spectral analysis

```
[y, fs0] = audioread("musique.wav\musique.wav")
```

```
y = 300000×1
    0.0001
    0.0002
    0.0002
    0.0000
   -0.0001
   -0.0001
   -0.0001
    0.0002
    0.0002
    0.0001
```

```
fs0 = 44100
```

```
plot(y)
```



Some constants:

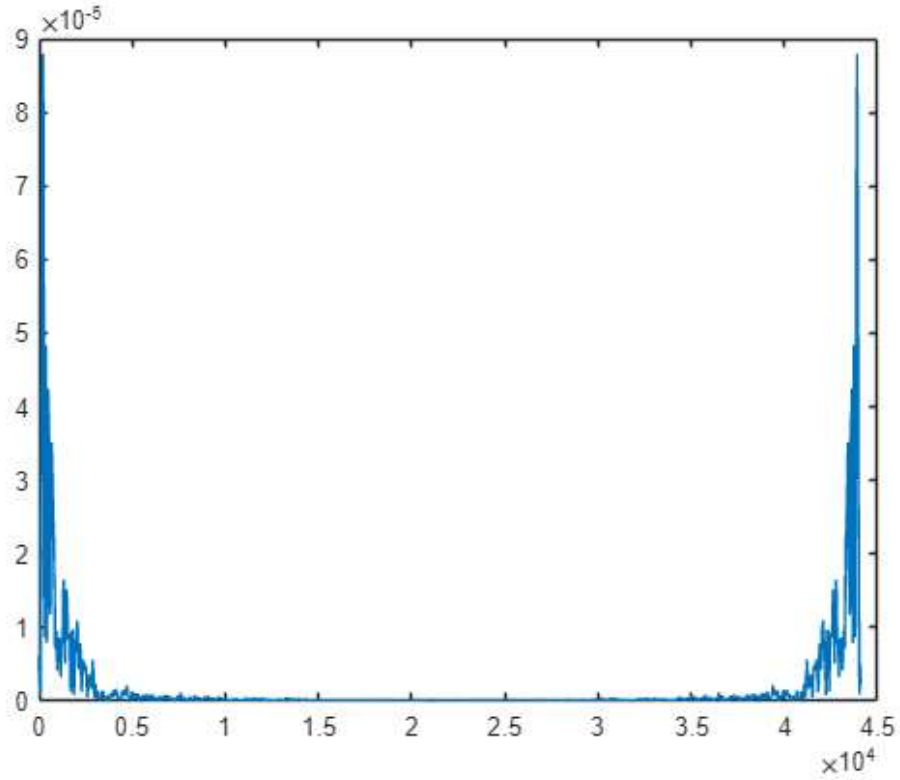
```
Nf = 4096;
fs = fs0;
```

The fourier transform:

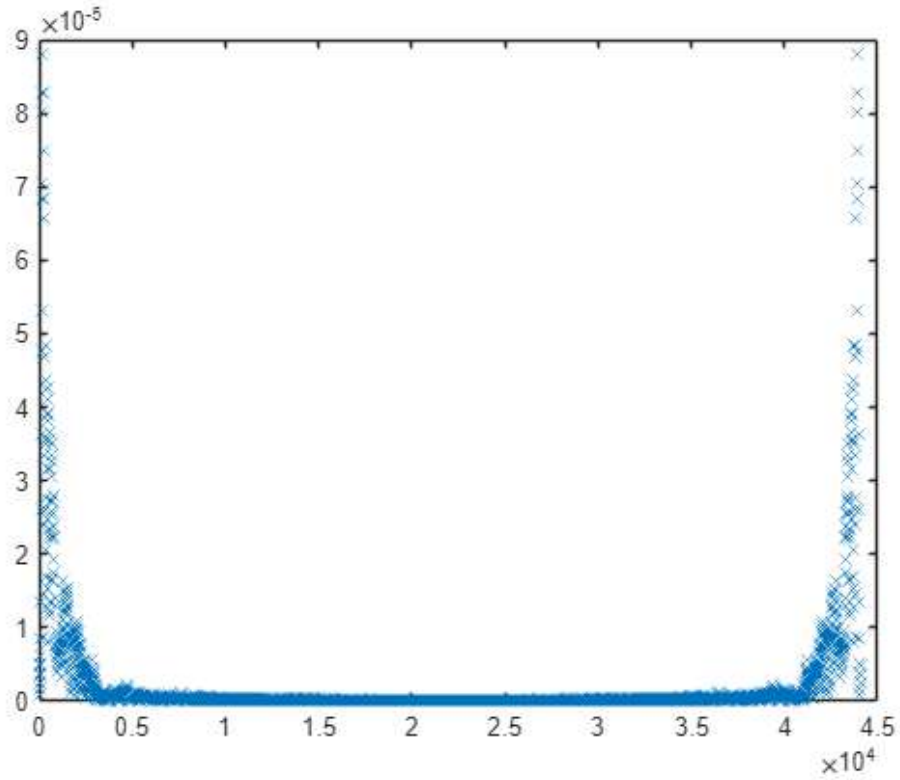
```
[f, tfx] = transffourier(y, Nf, fs);
```

The plot:

```
plot(f, abs(tfx))
```




```
plot(f, abs(tfx), 'x')
```



Peaks:

```
[pks, locs] = findpeaks(abs(tfx), f, 'MinPeakHeight',0.95*max(abs(tfx)))
```

```
pks = 2×1

10^-4 ×

    0.8796
    0.8796

locs = 1×2
```

```
10^4 ×
    0.0205    4.3895
```

C2.2 Coder-decoder

```
[y, Fs] = audioread("musique.wav\musique.wav")
```

```
y = 300000×1
    0.0001
    0.0002
    0.0002
    0.0000
   -0.0001
   -0.0001
   -0.0001
    0.0002
    0.0002
    0.0001


Fs = 44100
```

```
info = audioinfo("musique.wav\musique.wav")
```

```
info = struct with fields:
             Filename: '\\data-pfe\lgorecki2026\.Bureau\sigpro\C1and2and3\musique.wav\musique.wav'
    CompressionMethod: 'Uncompressed'
          NumChannels: 1
           SampleRate: 44100
         TotalSamples: 300000
             Duration: 6.8027
                Title: []
              Comment: []
               Artist: []
        BitsPerSample: 16
```

```
sound(y, Fs, info.BitsPerSample);
pause(length(y)/Fs + 0.5);
```

Encode

```
fmin = 0;
fmax = 8000;
Fs
```

```
Fs = 44100
```

```
[npt, scaling] = codeur(y, Fs, info.BitsPerSample,  fmin, fmax, "compressed\musique.wav");
```

Decode

```
[yd, fsd, bitsd] = decodeur("compressed\musique.wav", fmin, fmax, npt, scaling)
```

```
yd = 300000×1




   -0.0297
   -0.0038
```

```
    0.0113
    0.0122
    0.0052
   -0.0006
   -0.0008
    0.0023
    0.0042
    0.0031

fsd =  4100
bitsd = 16
```

Play

```
sound(yd, fsd, bitsd)
```